

EGCO 425 – Chapter 5

Classification Methods

K-Nearest Neighbors (KNN)

Naïve Bayes

References

- [1] Tan et al., Introduction to Data Mining (Chapters 4-5)
- [2] Han & Kamber, Data Mining: Concepts and Techniques (Chapters 8-9)
- [3] Kotu & Deshpande, Data Science: Concepts and Practice (Chapter 4)
- [4] RapidMiner Doc, <https://docs.rapidminer.com/latest/studio/operators/>

Chapter Outline

⊕ Proximity measures

▶ Distance

- Hamming
- Manhattan, Euclidean, Chebyshev

▶ Similarity

- Simple matching coefficient (SMC), Jaccard
- Cosine

⊕ K-nearest neighbors (KNN)

⊕ Naïve Bayes

- ▶ Conditional probability
- ▶ Laplace correction

Eager vs. Lazy Learners

⊕ Recall classification problem

- ▶ Learning phase : build classification model from training data
- ▶ Prediction phase : apply the model to testing or unseen data

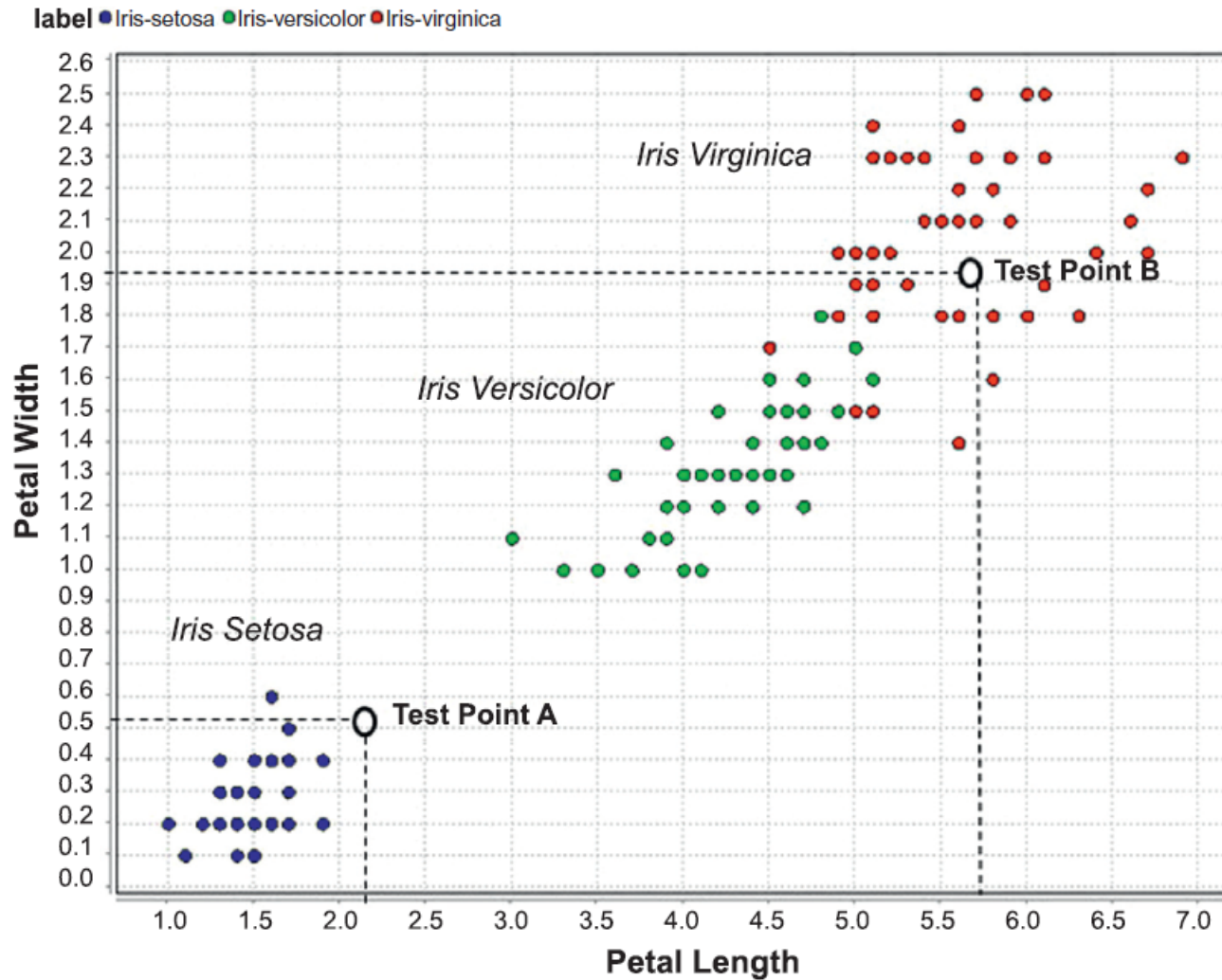
⊕ Eager learners

- ▶ Build model as soon as training data is available
- ▶ Decision trees, rule-based classifiers

⊕ Lazy learners

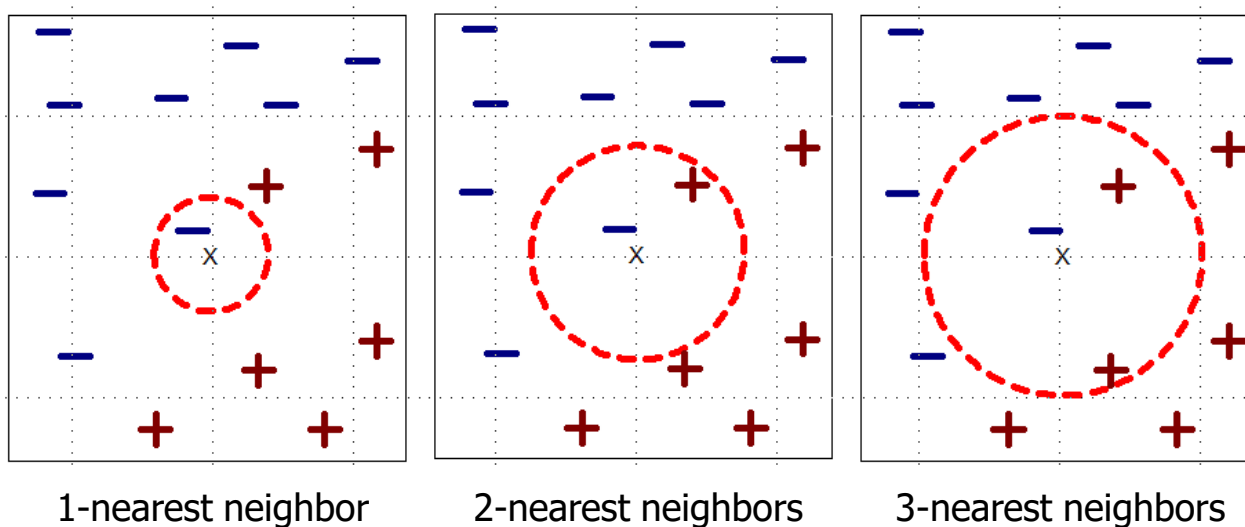
- ▶ Don't build any model, only keep training data
- ▶ Computation is done when predicting a new record
- ▶ In KNN, prediction is based on class labels of nearest training records

K-Nearest Neighbors (KNN)



⊕ Points to consider

- ▶ How to determine the distance between any 2 records ?
 - Attributes have different types e.g. nominal and numeric
 - Attributes have different scales e.g. age and salary
- ▶ How to determine the number of nearest neighbors (= K) ?
- ▶ How to determine the final prediction ?



Proximity Measures

⊕ Distance

- ▶ Between nominal values
- ▶ Between boolean values Hamming
- ▶ Between numeric values Manhattan, Euclidean, Chebyshev

⊕ Similarity

- ▶ Between nominal values
- ▶ Between boolean values simple matching coefficient, Jaccard
- ▶ Between numeric values cosine

Distance & similarity between nominal values

- ⊕ If 2 records are in the same category (same nominal value)
 - ▶ Distance = 0
 - ▶ Similarity = 1
- ⊕ If 2 records are in different categories (different nominal values)
 - ▶ Distance = 1
 - ▶ Similarity = 0
- ⊕ For example, attribute color = {red, green, blue}
 - ▶ Record 1's color = red
 - ▶ Record 2's color = green
 - ▶ Record 3's color = red
 - ▶ Between records 1 and 2 distance = 1 similarity = 0
 - ▶ Between records 1 and 3 distance = 0 similarity = 1

Distance & similarity between boolean values

⊕ Example : there are 10 bits (or booleans)

▶ Record 1 = 1 0 0 0 0 0 0 0 0 0

▶ Record 2 = 0 0 0 0 0 0 1 0 0 1

Hamming Distance	Simple matching coefficient (SMC) Similarity
$\frac{\text{total mismatches}}{\text{total bits}}$	$\frac{\text{total matches}}{\text{total bits}}$; $\text{SMC} = 1 - \text{Hamming}$
$3/10 = 0.3$	$7/10 = 0.7$

⊕ High SMC because of high co-absence is sometimes misleading

- ▶ For example, each attribute = buying a product (1=yes, 0=no)
- ▶ Knowing that both customers are in agreement to not buy most products in a supermarket is not much useful
- ▶ What about products both like to buy ?

⊕ Example : there are 10 bits (or booleans)

▶ Record 1 = 1 0 0 0 0 0 0 0 0 0

▶ Record 2 = 0 0 0 0 0 0 1 0 0 1

⊕ Jaccard coefficient $\frac{\text{total copresences}}{\text{total bits} - \text{total coabsences}} = 0/3 = 0$

⊕ Comparing SMC and Jaccard

▶ Use SMC when we give **equal weights** to both categories
E.g. congressmen voting **agree** or **disagree** on a number of issues

▶ Use Jaccard when we give **unequal weights** to both categories
E.g. customers **buying** a number of products

▶ Both can be extended to nominal values where
 $\text{matches} = \text{copresences} = \text{number of matched categories}$

Distance between numeric values

⊕ Definitions

- ▶ Records **X** has numeric values $(x_1, x_2, \dots, x_n)$
 - ▶ Records **Y** has numeric values $(y_1, y_2, \dots, y_n)$
- } **n** numeric attributes

⊕ **Manhattan distance** = $|x_1 - y_1| + |x_2 - y_2| + \dots + |x_n - y_n|$

- ▶ Or city-block, or taxicab distance
- ▶ Preferred over Euclidean distance when #attributes is very large

⊕ **Euclidean distance** = $\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}$

- ▶ Straight line distance between any 2 data points
- ▶ Most commonly used

⊕ **Chebyshev distance** = $\max \{ |x_1 - y_1|, |x_2 - y_2|, \dots, |x_n - y_n| \}$

⊕ Manhattan, Euclidean, Chebyshev are special cases of **Minkowski distance**

$$distance(X, Y) = \sqrt[r]{\sum_{i=1}^n (x_i - y_i)^r}$$

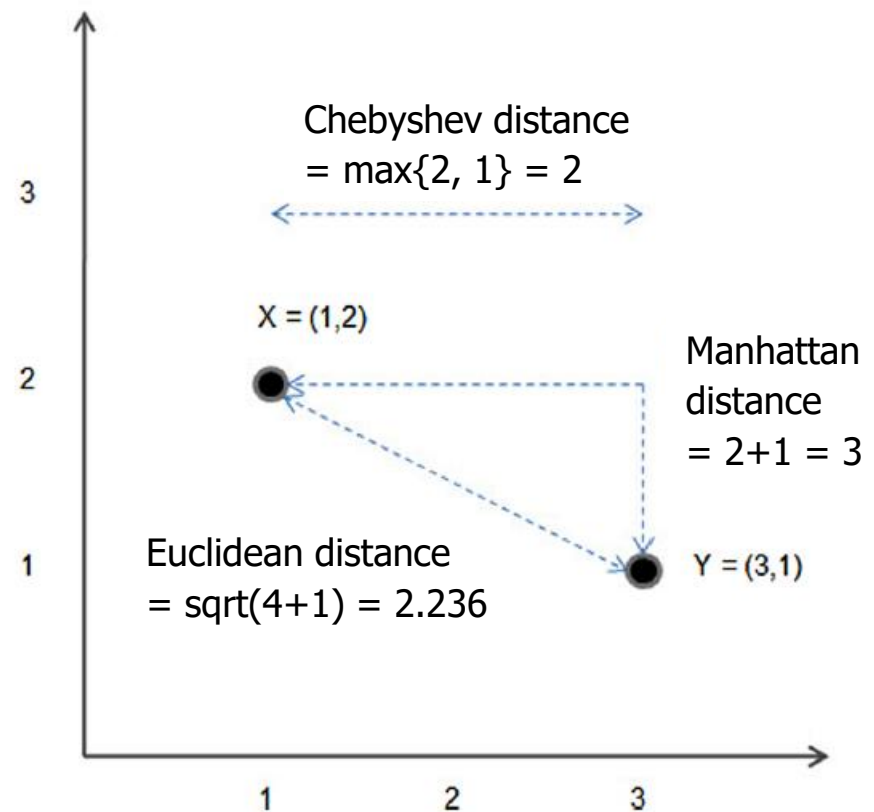
Manhattan $r = 1$

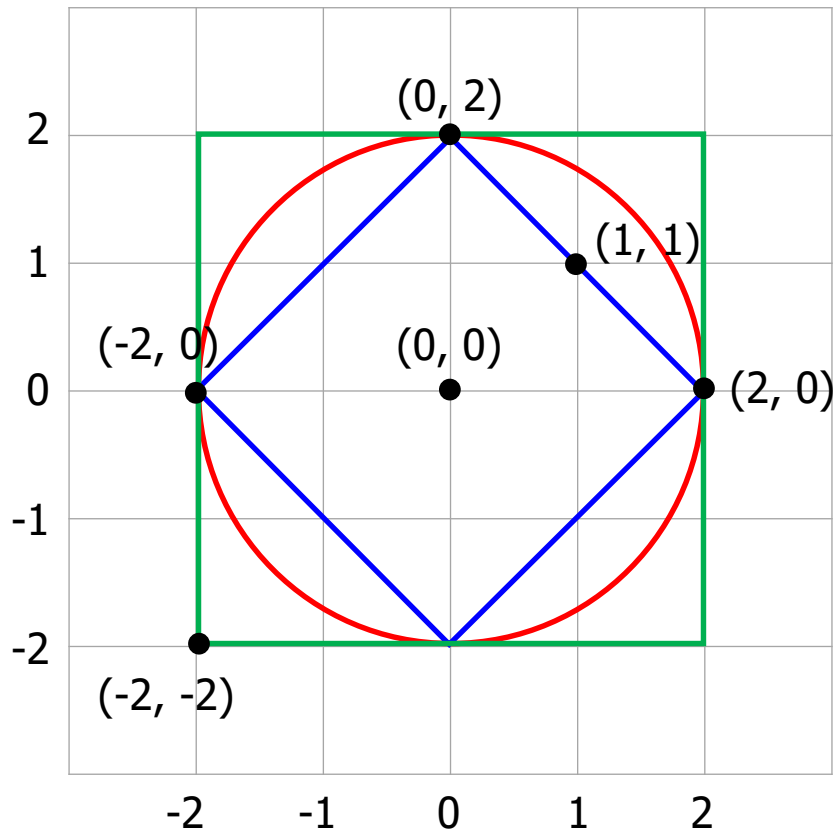
Euclidean $r = 2$

Chebyshev $r = \infty$

For Chebychev $\rightarrow$ apply $\lim_{r \rightarrow \infty}$
So distance

$$= \sqrt[r]{(\text{biggest diff})^r} = \text{biggest diff}$$





Radius of different distance functions

All points on red line have Euclidean distance = 2 from (0, 0)

All points on blue line have Manhattan distance = 2 from (0, 0)

All points on green line have Chebyshev distance = 2 from (0, 0)

Similarity between numeric values

Records **X** has numeric values $(x_1, x_2, \dots, x_n)$

Records **Y** has numeric values $(y_1, y_2, \dots, y_n)$

$$\oplus \text{ Cosine similarity} = \frac{X \cdot Y}{\|X\| \cdot \|Y\|} = \frac{\sum x_i y_i}{\left(\sqrt{\sum x_i^2}\right) \left(\sqrt{\sum y_i^2}\right)}$$

Dot product between 2 vectors

- ▶ Completely similar angle between vectors = 0°
- ▶ Completely dissimilar angle between vectors = 90°

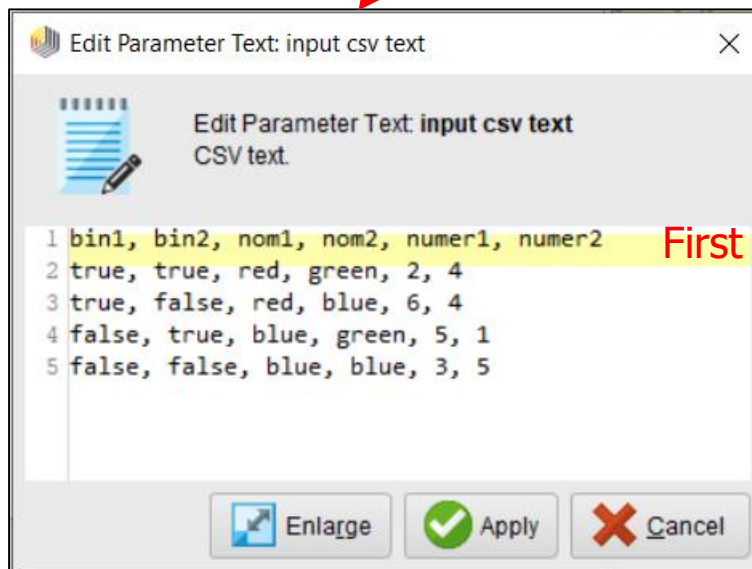
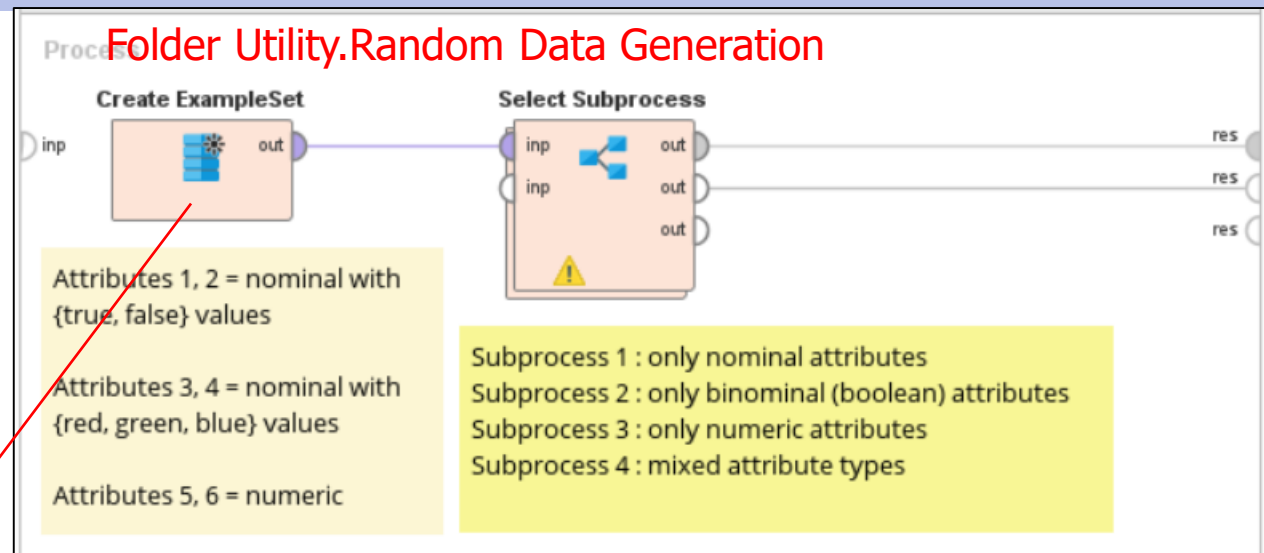
$\oplus$ Example

- ▶ $X = 5 \ 0 \ 3 \ 0 \ 2 \ 0 \ 0 \ 2 \ 0 \ 0$
- ▶ $Y = 3 \ 0 \ 2 \ 0 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1$
- ▶ Numerator = $15 + 6 + 2 + 2 = 25$
- ▶ Divisor = $\text{sqrt}(42) * \text{sqrt}(17) = 6.48 * 4.12$
- ▶ Cosine = $25 / 26.70 = 0.94$

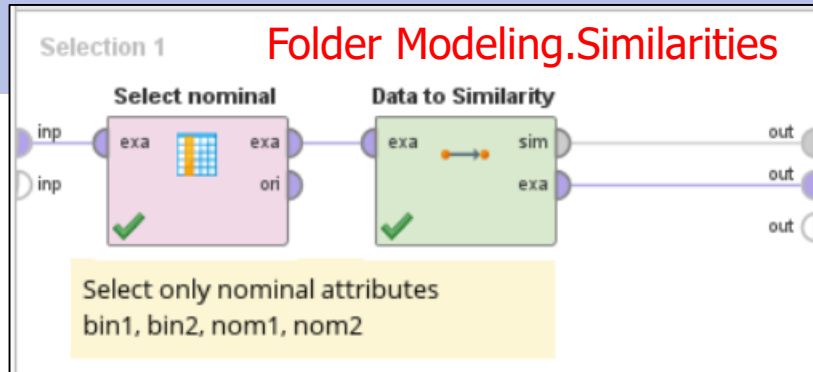
Issues in Proximity Calculation

- ⊕ A record consists of different types of attributes
- ⊕ Attributes have different scales
 - ▶ For example : Euclidean distance between people based on age and income attributes → distance is dominated by income
 - ▶ Fixed by normalizing each attributes
- ⊕ Attributes have different weights
 - ▶ Attributes contribute unequally to the proximity between objects
 - ▶ For example : distance between students should be based on their GPA rather than height
 - ▶ Fixed by attribute weighting → multiplying GPA by a weight factor
 - ▶ Use feature selection method to determine attribute scores

Example (1) : distance and similarity



First line = attribute names



Nominal

Row No.	id	bin1	bin2	nom1	nom2
1	1	true	true	red	green
2	2	true	false	red	blue
3	3	false	true	blue	green
4	4	false	false	blue	blue

{true, false} are treated as categories in nominal attributes. They are not boolean

First	Second	Distance
1.0	2.0	2
1.0	3.0	2
1.0	4.0	4
2.0	3.0	4
2.0	4.0	2
3.0	4.0	2

Pair of records

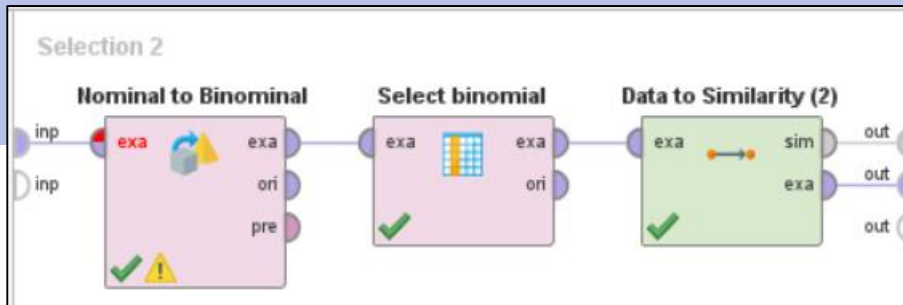
Nominal
distance

First	Second	Similarity
1.0	2.0	0.500
1.0	3.0	0.500
1.0	4.0	0
2.0	3.0	0
2.0	4.0	0.500
3.0	4.0	0.500

= 2/4

Jaccard treats each
match as co-presence
(no co-absence)

Same scores for SMC and Jaccard
because for nominal attributes, similarity = $\frac{\#matches}{\#attributes}$



{true, false} are now **boolean** values of binominal attributes. Jaccard treats true = presence, false = absence

Row No.	id	bin1 = true	bin1 = false	bin2 = true	bin2 = false	nom1 = red	nom1 = blue	nom2 = green	nom2 = blue
1	1	true	false	true	false	true	false	true	false
2	2	true	false	false	true	true	false	false	true
3	3	false	true	true	false	false	true	true	false
4	4	false	true	false	true	false	true	false	true

First	Second	Distance
1.0	2.0	4
1.0	3.0	4
1.0	4.0	8
2.0	3.0	8
2.0	4.0	4
3.0	4.0	4

Nominal distance

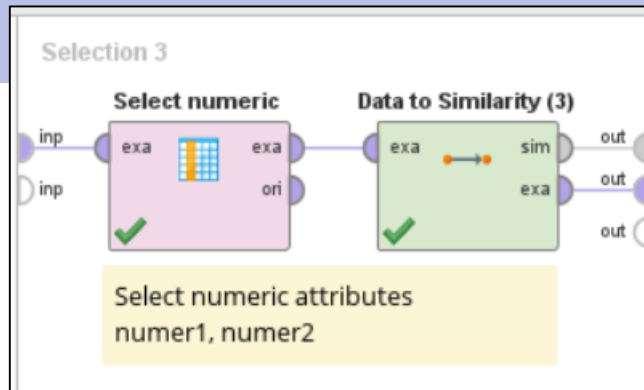
First	Second	Similarity
1.0	2.0	0.500 = 4/8
1.0	3.0	0.500
1.0	4.0	0
2.0	3.0	0
2.0	4.0	0.500
3.0	4.0	0.500

SMC = #matches / 8

First	Second	Similarity
1.0	2.0	0.333 = 2/6
1.0	3.0	0.333
1.0	4.0	0
2.0	3.0	0
2.0	4.0	0.333
3.0	4.0	0.333

Jaccard

$$= \text{\#both true} / (\text{\#both true} + \text{\#both false})$$



Row No.	id	numer1	numer2
1	1	2	4
2	2	6	4
3	3	5	1
4	4	3	5

Numeric

First	Second	Distance
1.0	2.0	4
1.0	3.0	4.243
1.0	4.0	1.414
2.0	3.0	3.162
2.0	4.0	3.162
3.0	4.0	4.472

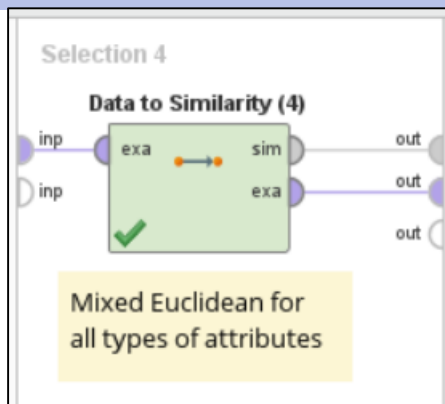
Euclidean distance

First	Second	Similarity
1.0	2.0	-2.333
1.0	3.0	-7
1.0	4.0	-2.167
2.0	3.0	-1.889
2.0	4.0	-1.900
3.0	4.0	-3.333

Jaccard for numeric values

$$\frac{12 + 16 = 28}{8 + 8 - 28} = \frac{28}{-12}$$

$$Jaccard = \frac{\sum x_i y_i}{(\sum x_i) + (\sum y_i) - (\sum x_i y_i)}$$



Row No.	id	bin1	bin2	nom1	nom2	numer1	numer2
1	1	true	true	red	green	2	4
2	2	true	false	red	blue	6	4
3	3	false	true	blue	green	5	1
4	4	false	false	blue	blue	3	5

Mixed types

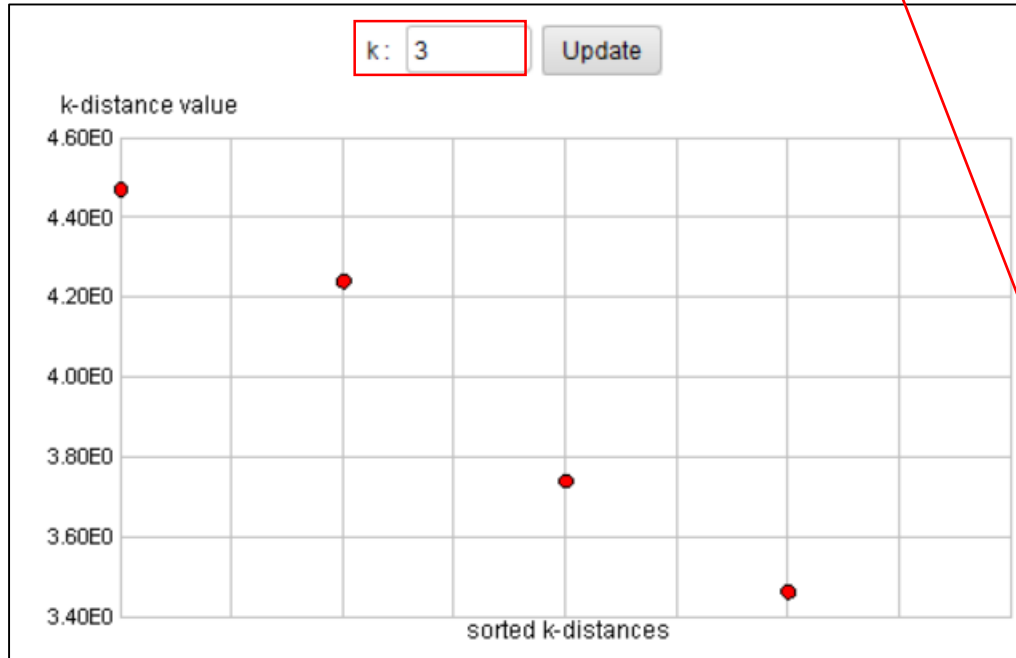
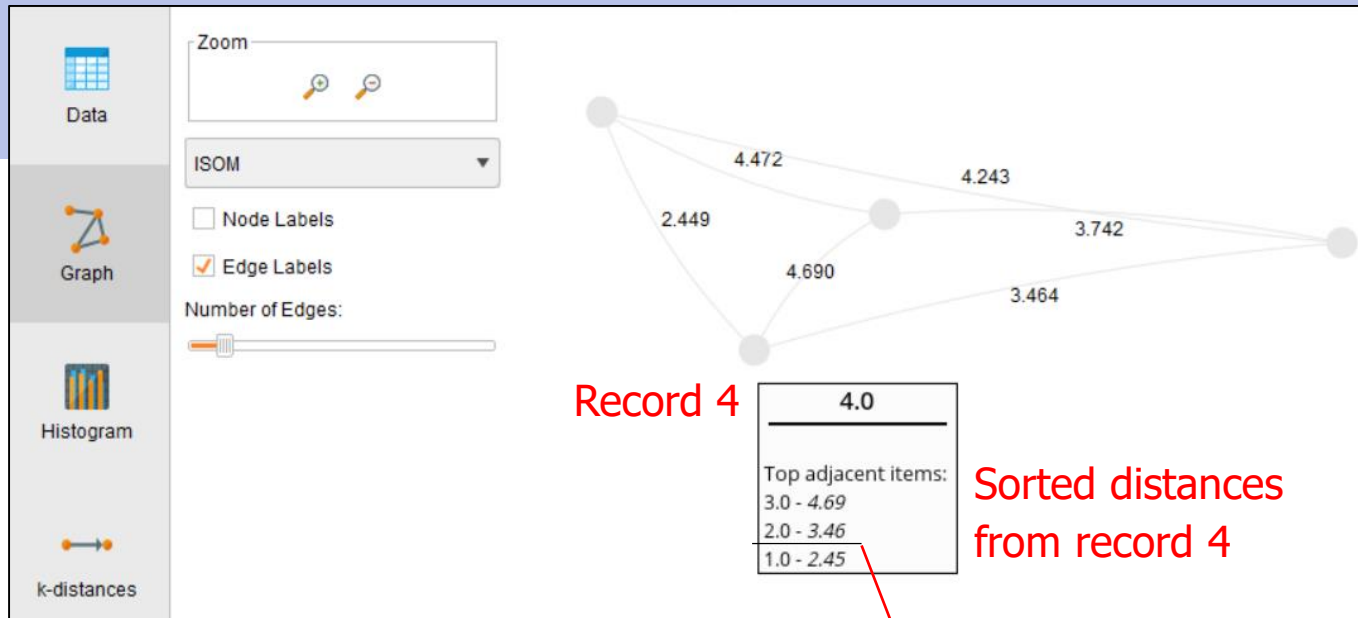
$$\sqrt{(2) + (4^2)} = \sqrt{18}$$

First	Second	Distance
1.0	2.0	4.243
1.0	3.0	4.472
1.0	4.0	2.449
2.0	3.0	3.742
2.0	4.0	3.464
3.0	4.0	4.690

$$\text{Mixed Euclidean} = \sqrt{\sum d_{\text{nominal}} + \sum d_{\text{numeric}}}$$

d_{numeric} is calculated as in Euclidean function

$d_{\text{nominal}} = 1$ for different categories



Distances to 2nd nearest neighbor of all records ($k = 3$)

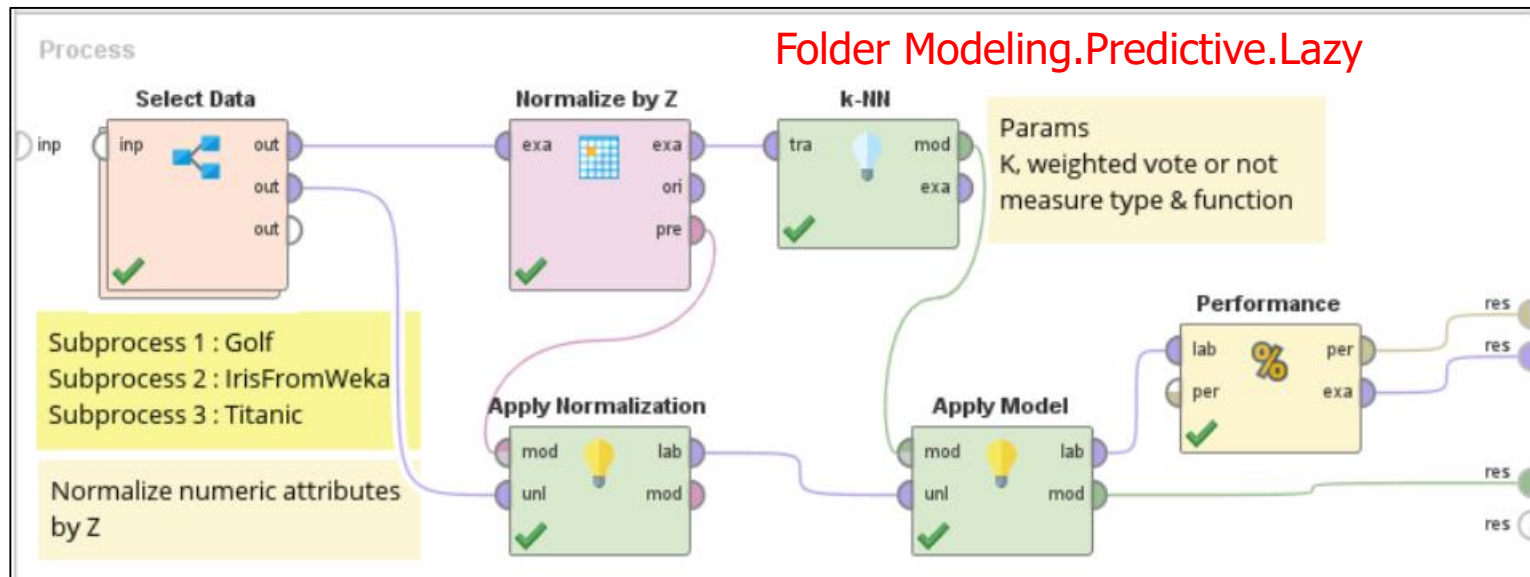
For record 2, distance = 4.47

For record 1, distance = 4.24

For record 3, distance = 3.74

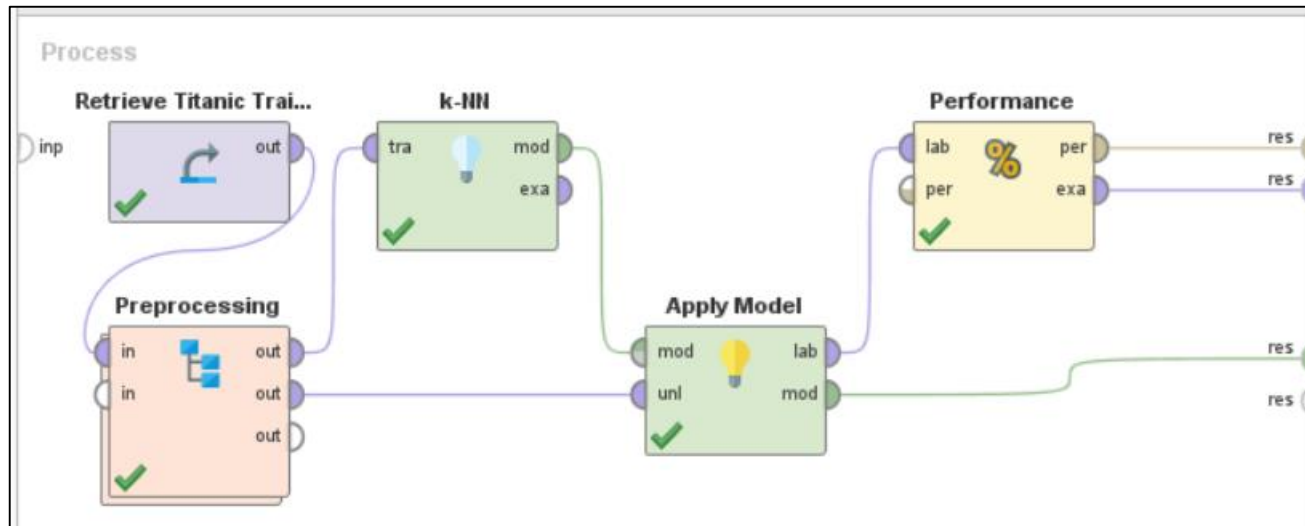
For record 4, distance = 3.46

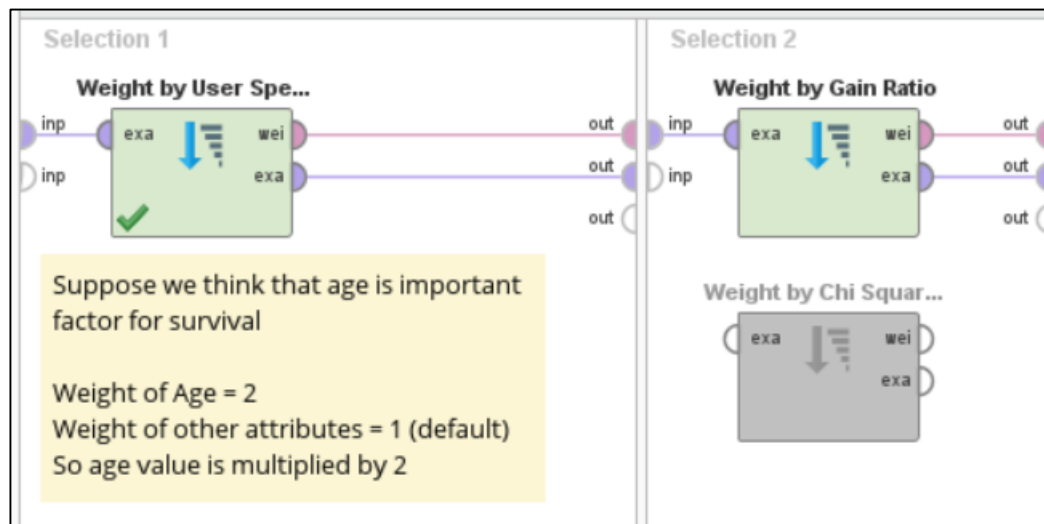
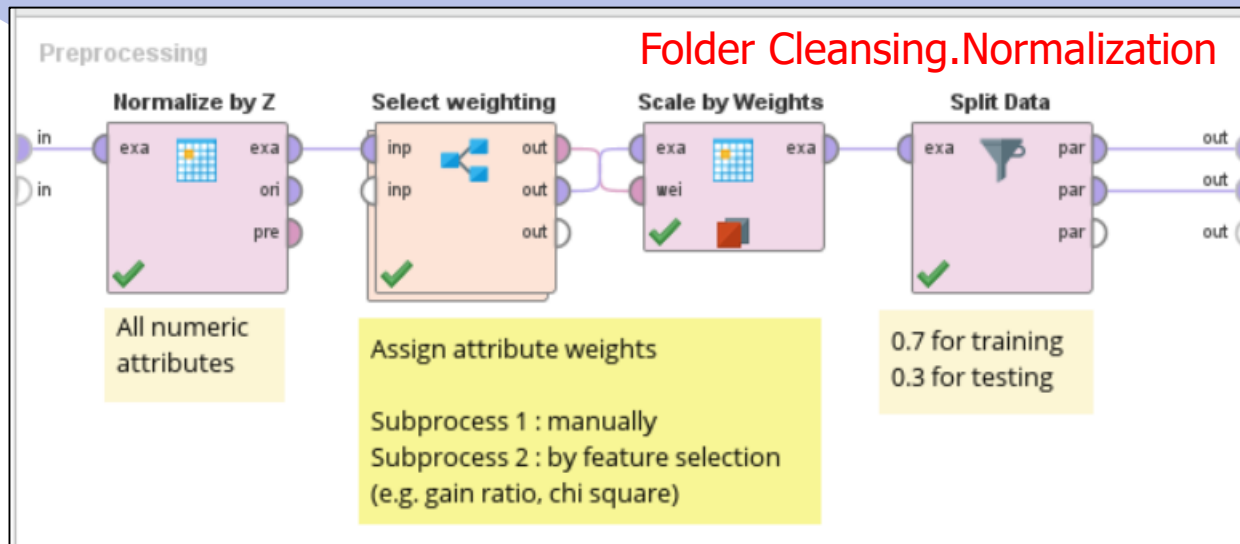
Example (2) : KNN



- ⌘ If numeric attributes have different scales → normalize them, so that attributes with big scales don't dominate distance calculation
- ⌘ Odd K value is recommended to avoid tie. Each neighbor's vote can be weighted by the distance between that neighbor and target

Example (3) : KNN with attribute weighting





Folder Modeling.Feature Weights

KNN Summary

- ⊕ No learning phase
 - ▶ No model, less descriptive power
 - ▶ Don't take into account the relevancy or predictive power of each attribute → require manual preprocessing e.g. normalization, feature selection, attribute weighting
- ⊕ Prediction is expensive, especially if training data is very big, because we have to compare target record with every training record to find K nearest neighbors
- ⊕ Prediction is based on local information i.e. only K records
 - ▶ K is too small → sensitive to noise, overfitting
 - ▶ K is too big → misclassification (dissimilar records are included)
- ⊕ KNN is a popular method to impute missing values (see Chapter 3)

Naïve Bayes

⊕ Definitions

- ▶ $X = (A_1, A_2, \dots, A_n)$ is a set of attributes
- ▶ Y is class (yes or no) that we want to predict

⊕ Prediction is done by comparing $P(Y=\text{yes} | X)$ and $P(Y=\text{no} | X)$ and choose class label with higher probability

⊕ From Bayes Theorem

$$P(\text{yes} | X) = \frac{P(X | \text{yes}) P(\text{yes})}{P(X)} \quad ; \quad P(\text{no} | X) = \frac{P(X | \text{no}) P(\text{no})}{P(X)}$$

- ▶ $P(X)$ can be ignored
- ▶ But finding $P(X | \text{yes})$, $P(X | \text{no})$ is not easy because X is not a single value → we have to compute $P(A_1, A_2, \dots, A_n | \text{yes})$
 $P(A_1, A_2, \dots, A_n | \text{no})$

Naïve Assumption

- ⊕ Assuming that attributes are independent

$$P(A_1, A_2, \dots, A_n \mid \text{yes}) = \prod_{i=1}^n P(A_i \mid \text{yes})$$

- ⊕ To compute each $P(A_i = a \mid \text{yes})$ from training data
 - ▶ Nominal attribute : use relative frequency that $A_i = a$ in class yes
 - ▶ Numeric attribute : use probability density from Normal distribution function

$$P(A_i = a \mid \text{yes}) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(a-\mu)^2}{2\sigma^2}}$$

μ and σ = mean and SD of A_i in class yes

Excel function = `NORM.DIST(a,  $\mu$ ,  $\sigma$ , false)`

Not quite correct in stat theory, but we do it for rough estimation

From training data

TID	House	Marital Status	Income	Default
1	owner	single	125K	no
2	tenant	married	100K	no
3	tenant	single	70K	no
4	owner	married	120K	no
5	tenant	divorced	95K	yes
6	tenant	married	60K	no
7	owner	divorced	220K	no
8	tenant	single	85K	yes
9	tenant	married	75K	no
10	tenant	single	90K	yes

- $P(\text{owner} \mid \text{yes}) = 0/3$
 $P(\text{tenant} \mid \text{yes}) = 3/3$
- $P(\text{owner} \mid \text{no}) = 3/7$
 $P(\text{tenant} \mid \text{no}) = 4/7$
- $P(\text{single} \mid \text{yes}) = 2/3$
 $P(\text{married} \mid \text{yes}) = 0/3$
 $P(\text{divorced} \mid \text{yes}) = 1/3$
- $P(\text{single} \mid \text{no}) = 2/7$
 $P(\text{married} \mid \text{no}) = 4/7$
 $P(\text{divorced} \mid \text{no}) = 1/7$
- For income
 no $\rightarrow$ mean = 110, variance = 2975
 yes $\rightarrow$ mean = 90, variance = 25

Predicting a new record

- ⊕ New record $X = (\text{home}=\text{tenant}, \text{marital}=\text{married}, \text{income}=100\text{K})$
- ⊕ $P(\text{yes} \mid X) = P(X \mid \text{yes}) P(\text{yes}) / P(X)$

$$= P(\text{tenant} \mid \text{yes}) P(\text{married} \mid \text{yes}) P(100\text{K} \mid \text{yes}) P(\text{yes})$$

$$= (3/3)(0/3)(0.0108)(3/10)$$

$$= 0$$
- ⊕ $P(\text{no} \mid X) = P(X \mid \text{no}) P(\text{no}) / P(X)$

$$= P(\text{tenant} \mid \text{no}) P(\text{married} \mid \text{no}) P(100\text{K} \mid \text{no}) P(\text{no})$$

$$= (4/7)(4/7)(0.0072)(7/10)$$

$$= 0.0016$$
- ▶ Normalize each probability by $P(\text{yes} \mid X) + P(\text{no} \mid X) = 0.0016$, so they sum to 1 → $P(\text{yes} \mid X) = 0, P(\text{no} \mid X) = 100\%$
- ▶ Prediction = no

Laplace Correction

- ⊕ Zero prob. in one term causes the whole $P(\text{yes} \mid X)$ to be zero
- ⊕ To avoid this case, we add 1 to the frequency of each category
If attribute A has k categories
 - ▶ Laplace corrected probability

$$P(A = a \mid \text{yes}) = \frac{\text{number of records in category } a + 1}{\text{number of records in yes} + k}$$

	<u>Old</u>	<u>New</u>
• $P(\text{owner} \mid \text{yes})$	= 0/3	1/5
$P(\text{tenant} \mid \text{yes})$	= 3/3	4/5
• $P(\text{owner} \mid \text{no})$	= 3/7	4/9
$P(\text{tenant} \mid \text{no})$	= 4/7	5/9

	<u>Old</u>	<u>New</u>
• $P(\text{single} \mid \text{yes})$	= 2/3	3/6
$P(\text{married} \mid \text{yes})$	= 0/3	1/6
$P(\text{divorced} \mid \text{yes})$	= 1/3	2/6
• $P(\text{single} \mid \text{no})$	= 2/7	3/10
$P(\text{married} \mid \text{no})$	= 4/7	5/10
$P(\text{divorced} \mid \text{no})$	= 1/7	2/10

Predicting a new record (Laplace corrected)

- ⌘ New record $X = (\text{home}=\text{tenant}, \text{marital}=\text{married}, \text{income}=100\text{K})$
- ⌘ $P(\text{yes} \mid X) = P(X \mid \text{yes}) P(\text{yes}) / P(X)$

$$= P(\text{tenant} \mid \text{yes}) P(\text{married} \mid \text{yes}) P(100\text{K} \mid \text{yes}) P(\text{yes})$$

$$= (4/5)(1/6)(0.0108)(3/10)$$

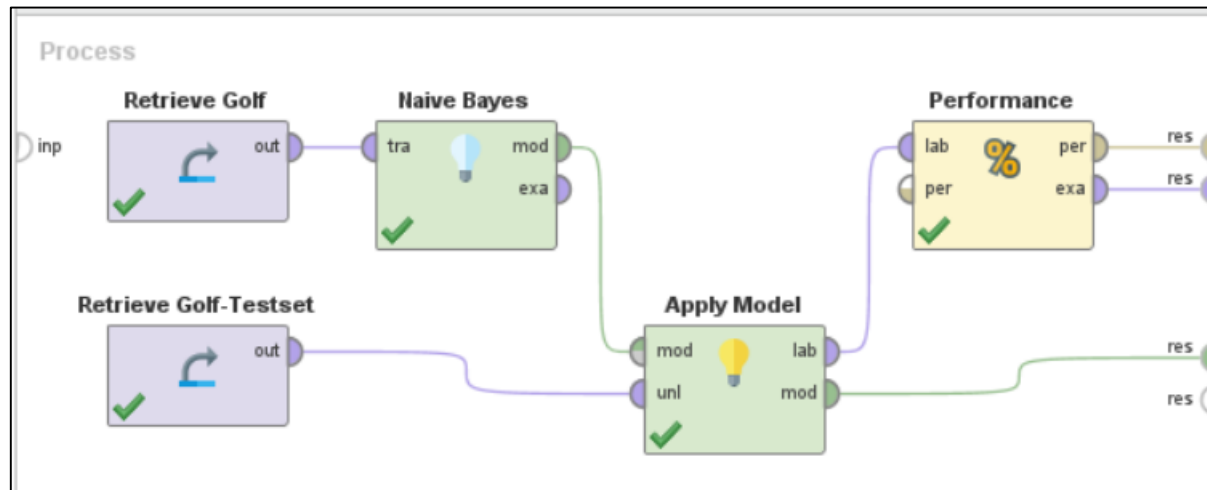
$$= 0.000432$$
- ⌘ $P(\text{no} \mid X) = P(X \mid \text{no}) P(\text{no}) / P(X)$

$$= P(\text{tenant} \mid \text{no}) P(\text{married} \mid \text{no}) P(100\text{K} \mid \text{no}) P(\text{no})$$

$$= (5/9)(5/10)(0.0072)(7/10)$$

$$= 0.0014$$
- ▶ Normalize each probability by $P(\text{yes} \mid X) + P(\text{no} \mid X) = 0.001832$, so they sum to 1 $\rightarrow P(\text{yes} \mid X) = 23.58\%$, $P(\text{no} \mid X) = 76.42\%$
- ▶ Prediction = no

Example (4) : Naïve Bayes



Row No.	Play	prediction(Play)	confidence(no)	confidence(yes)	Outlook	Temperature	Humidity	Wind
1	yes	no	0.711	0.289	sunny	85	85	false
2	no	yes	0.058	0.942	overcast	80	90	true
3	yes	yes	0.014	0.986	overcast	83	78	false
4	yes	yes	0.412	0.588	rain	70	96	false
5	yes	yes	0.460	0.540	rain	68	80	true



Description



Simple Charts



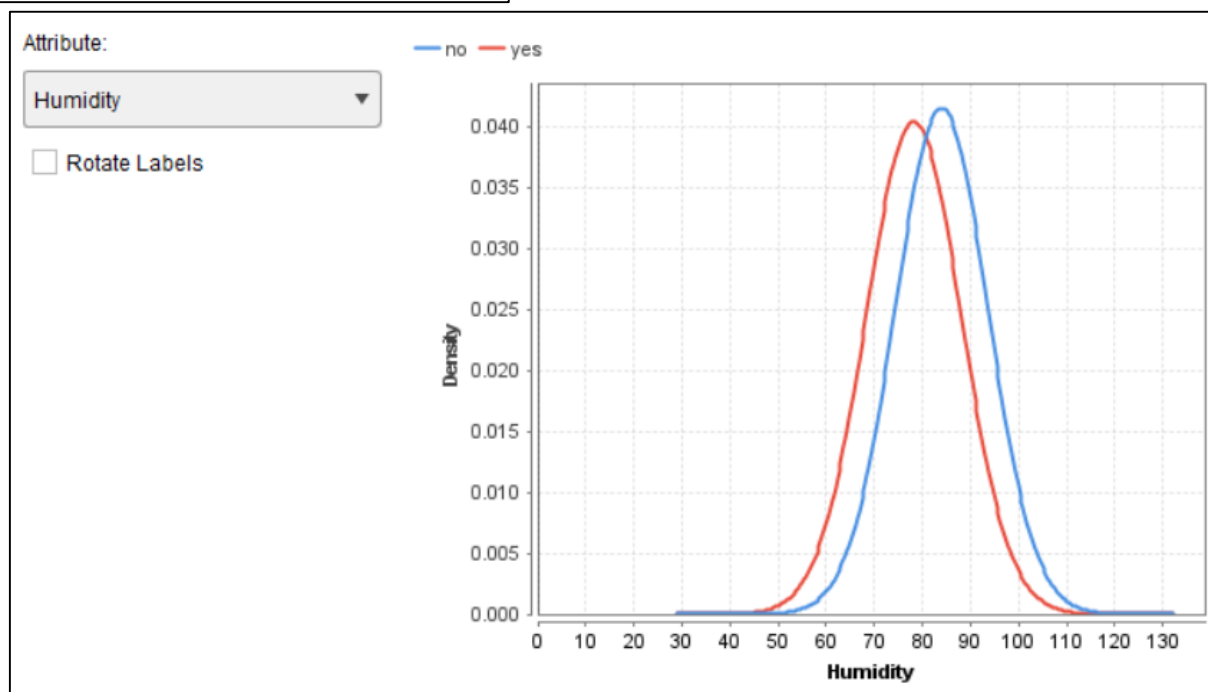
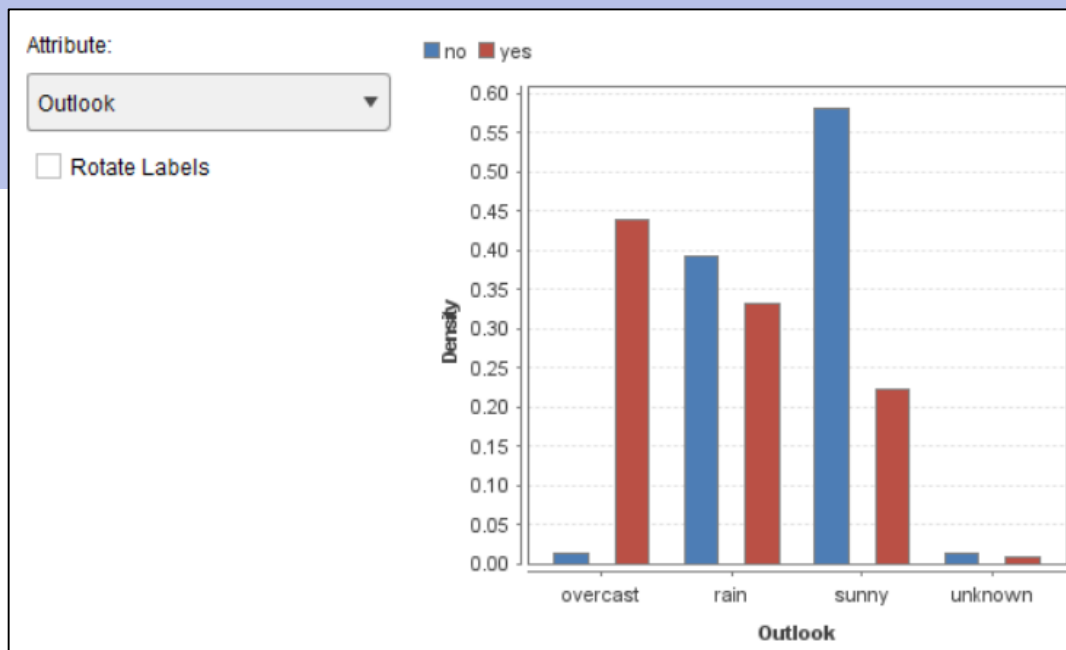
Distribution Table



Annotations

Attribute	Parameter	no	yes
Outlook	value=rain	0.392	0.331
Outlook	value=overcast	0.014	0.438
Outlook	value=sunny	0.581	0.223
Outlook	value=unknown	0.014	0.008
Temperature	mean	74.600	73
Temperature	standard deviation	7.893	6.164
Humidity	mean	84	78.222
Humidity	standard deviation	9.618	9.884
Wind	value=true	0.589	0.333
Wind	value=false	0.397	0.659
Wind	value=unknown	0.014	0.008

For nominal attribute, unknown category is added (without Laplace correction, its frequency = 0)



Naïve Bayes Summary

- ⊕ Naturally handle noises and missing values
 - ▶ When estimating conditional probability, noises are averaged out & missing values are ignored (not included in the calculation)
- ⊕ Robust to irrelevant attributes
 - ▶ Irrelevant attribute tends to have almost unique values or identical values for all records, hence uniform distribution
 - ▶ So its conditional prob. has no impact on the overall prob.
- ⊕ Important assumption = attributes are independent of each other
 - ▶ Difficult to prove in practice
 - ▶ In many cases, it works well despite the assumption being violated